On the Formal Inexplicability of Self-Evident Metaphysical Phenomena and Related Systems

Abstract. Since Descartes first proclaimed "cogito, ergo sum" back in 1637, "I think, therefore I am" has become the Declaration of Independence for consciousness. Chalmers' informed us of the fact that science has a constitutionally *Hard Problem* on their hands. If he's as right as we believe he is, conscious experience may forever elude our most powerful explanatory framework for physical phenomena. In other news, a group referring to themselves as the *IIT-Concerned* recently proclaimed in Nature Neuroscience that any theory of consciousness that can't stand up to the scrutiny of science shall be deemed "unscientific." And the emergence of Semantic AI has transformed this philosophical oddity into a practical concern with real ethical implications. We simply cannot know whether AI systems are conscious without first knowing the necessary and sufficient conditions for consciousness. How can we provide a formal definition of conscious experience that preserves the essence of its metaphysical character in a framework that is physically falsifiable?

Keywords: Formal Theories of Consciousness · The Hard Problem · Subjectivity · Semantic AI · Metaphysical Transduction.

1 A Colloquial Derivation

Almost 2/3 of surveyed philosophers acknowledge the existence of the Hard Problem, and 1/3 of them think that it's actually hard! [3] Consciousness is a field that spans disciplines, there is nothing that resemble scientific consensus, and there is nothing that resembles a testable theory. But we do have a compellingly elegant framing of the problem. And his framing of the problem is real enough to be *real* by majority vote. So we shall begin with Chalmers.

In his seminal work "Facing Up to the Problem of Consciousness," David Chalmers begins with a powerful observation that establishes the paradoxical nature of our subject: "There is nothing that we know more intimately than **conscious experience**, but there is nothing that is harder to explain." From the first part of this statement, we can immediately identify a fundamental characteristic of conscious experience that echos with Descartes: it is a **self-evident** phenomenon. Nothing is known more intimately than conscious experiences. They present themselves directly to their observer, requiring no justification beyond the experience itself.

The second part of Chalmers' statement suggests difficulty of explanation, but he goes on to clarify just how profound this difficulty is. Chalmers argues that "the emergence of experience goes beyond what can be derived from physical theory" and "the facts about experience cannot be an automatic consequence of any physical account." These statements establish that consciousness isn't merely difficult to explain within our current scientific framework but fundamentally resistant to such explanation. It is **inexplicable** in principle He asks

pointedly: "Why should physical processing give rise to a rich inner life at all? It seems objectively unreasonable that it should, and yet it does."

Regarding the relationship between conscious experience and physical processes, Chalmers makes a crucial observation: "Experience may arise from the physical, but it is not entailed by the physical." This statement positions consciousness in a unique relationship to physical reality. According to Merriam-Webster, "metaphysical" refers to that which relates to "the transcendent or to a reality beyond what is perceptible to the senses." Consciousness precisely fits this definition: it transcends physical explanation while still relating to physical processes. It is fundamentally **metaphysical** in nature.

Throughout his analysis, Chalmers treats consciousness as a **phenomenon**, both in the sense of phenomenal experience and as an object of scientific study. The term "phenomenon" captures this dual aspect perfectly. A phenomenon is both "an object or aspect known through the senses" and "a fact or event of scientific interest susceptible to scientific description and explanation." Consciousness is simultaneously the subjective experience itself and the object of our scientific inquiry. The etymological connection between "phenomenon" and "phenomenal" reinforces this as the appropriate term for our work.

By synthesizing these insights from Chalmers, we can formulate a definition that captures the essential nature of a conscious experience:

[Colloquial] Definition 1. *A Conscious Experience is an inexplicable yet self-evident metaphysical phenomenon.*

[Colloquial] Example 1. *I catch you admiring a rose in my garden and I absentmindedly ask "how did you figure out its color?" You chuckle and reply "It's red, Doug. I promise." Why do I feel like everyone always doubts my sanity when I ask them that question?*

2 The Semantics of *Colloquialism*

The *colloquial* refers to language that is familiar, conversational, and notably *informal*. Newton famously employed the use of colloquialism to pave the way to the mathematical phrasing of his Laws of Motion. And colloquialism compounds! "For every action there is an equal and opposite reaction" is itself the colloquial form of Newton's Third Law of Motion. The original is actually written in Latin. The canonical translation reads:

> To every action, there is always opposed an equal reaction; or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts. [15]

His colloquial phrasings serve as the informal counterpart to lengthy formal definitions, and his purpose was to imbue the syntactic formalism with a sense of meaning that was more relatable. In our exploration of *inexplicability*, we will encounter a friction between what can be entailed semantically and what can

be derived formally. The semantic form carries essential meaning in this case. In this spirit, we will replace the word "colloquial" with "semantic" going forward, and recognize the distinction between a *Semantic* form and a *Formal* form, and use the modern computational tools of reference to present both forms:

[Semantic] Definition 1. *A Conscious Experience is an inexplicable yet self-evident metaphysical phenomenon [1].*

3 A Self-Evident Perspective

There is an artful ambiguity in our definition of a conscious experience that might not be self-evident at a glance. In what *sense* do we mean "self-evident"? Self-evident to whom?

Insofar as an axiom of formal logic captures an aspect of *self-evident* in a formal sense, and it can be seen right alongside *inexplicable* in the words of Aristotle:

> We must know the primary things by themselves; for they are indemonstrable, and must be known by intuition — Aristotle, *Posterior Analytics*

Note, however that he uses the phrase "we" as a declaration of objectivity. When you and I read that phrase, he means we each should take the axioms of logic as self-evident, whatever they may be.

Kant leaves "we" implicit in his declaration of this ponderously self-evident truth:

> It is self-evident that the whole is greater than its part — Immanuel Kant, *Critique of Pure Reason*, 1781

If it is indeed self-evident to me, it is evident that it is qualitative rather than quantitative, because addition is the operator used to equate a whole number to its parts. But must it apply to that which is represented by that number? Perhaps. Either way, we *the reader* are expected to find Kant's proposition self-evident, and not some other hypothetical third-party.

The Founding Fathers of the United States made wonderful use of self-evident to lend identity to a nation.

> We hold these truths to be self-evident, that all men are created equal — Thomas Jefferson, *Declaration of Independence*, 1776

I personally find this to be the quintessential use of the phrase "self'evident" and the one that best expresses its use in this theory. Observe that by taking Kant and Jefferson together, you get an inclusive sense of identity for all mankind as a cohesive whole! And the whole constituency is constituted within a framework that's capable and has been proven capable of extending its definition of the word "men" to include "women" too. A woman's right to vote was once inexplicable

within a formal framework, and it is now self-evident by way of an amendment to the constitution: a new axiom which essentially redefines:

men => men & women

In the context of a conscious experience, "inexplicable yet self-evident" is meant in the *sense* that it apparent only and uniquely to the subject of the conscious experience.

So in our emerging formalism, the sense of *self* in "self-evident" is thing you hyperbolically declare that you lost when you say "I do not feel like my*self* today." So as the subject of that proposition, you must possess one. And to say "I simply must *see* it for my*self*" is a testament to irreplacable form of *evidence* that can only be provided by a conscious experience.

In all phrases in which the *self* is expressed, "I" is used to denote the identity of that which possesses it. This is the essence of the first-person perspective. Linguistic constructs like "I feel like my*self* again" reveal that the *identity* is the subject and the *self* is its object. We can use this intuition to formulate a semantic assertion:

[Semantic] Assertion 1. *A* Self *is a unique possession of the subject of a conscious experience. An* Identity *is that which possesses one. [1].*

4 What Qualifies as a Coherent Perspective?

The notion of a *quality* appears in the literature as a sort of perceptual equivalent to the type of property that an object might possess, in an objective sense. An rose has color as a property, as defined the characteristics of the photons that reflect off of it, and it has various physical properties that can be quantifiable. And there is the rose that we percieve which is red. it has the quality of red. Philosophers will refer to the instantaneous experience of a red as a "quale", and the collection of these instantaneous experiences for a given quality are its "qualia." It's something like that.

Chalmers does not introduce the phrase in his presentation of the hard problem in the "Facing Up to the Problem of Consciousness" so I will avoid it. But I will make use of the notion of *quality* rather than *property* when identifying how things *look* within the perspective afforded by a conscious experience.

However we choose to refer to the qualities we perceive, and the identity of the objects we perceive to have them, Chalmers gives us a thorough exposé of we experience them:

> When we see, for example, we experience visual sensations: the felt quality of redness, the experience of dark and light, the quality of depth in a visual field. Other experiences go along with perception in different modalities: the sound of a clarinet, the smell of mothballs. Then there are bodily sensations, from pains to orgasms; mental images that are conjured up internally; the felt quality of emotion, and the experience of a stream of conscious thought. What unites all of these states is that

> there is something it is like to be in them. All of them are states of experience.

Note that he remarks, with thoughtful philosophical phrasing, about "the felt quality of redness" so as to suggest that a color and a feeling share a commonality as a sort of irreducible or axiomatic quality. Let us use *objective perspective* as a term to denote that we denote the objects that we experience as "real" and the qualities we perceive correspond to something physically measurable. From the physical perspective, we *see* a red rose. We certainly experience *redness* somehow, but can we semantically justify the act of *seeing* an individual color as a distinct perceptual object?

Observe what examples language has to offer. To "see red" is a metaphorical idiom in English that artfully employs synesthesia to convey the idea that one's anger is so *real* that it has come to take on the quality of a physical object: color. Even in metaphorical form, red is still has the status of a quality.

We also have the ability to perceive red in conceptual form, where we objectify in the sense that we provide it with "red" as an identity, it so that it too can possess qualities. A particular shade of red might be *vivid* or *sensational* in relation to another shade or some canonical subjective shade that we each possess. But you'll note that "redness" is not actually one of the qualities that the objective form of "red" actually possesses. "The redness of red" is a phrase that expresses its inexplicable yet self-evident ability to serve both as a quality of an object that lends meaning to its identity, while also possessing a form of identity itsself. Does an identity possess the quality of being itself? Do I admit the "redness of red" and "Dougness of Doug" into my vocabulary?

We explored the notion of "self" above, and one might say that "Dougness" is a third-person synonym for the thing that I possess which I refer to as a "self" in the first-person perspective. But I earned that perspective by qualifying for it. And I qualify for it by having a sense of perspective. As I possess a perspective of my own, I do not find "Dougness" to be a semantically useful distinction. I don't have to qualify my "Dougness" to anyone. I am Doug!

To say the color "red" has a sense of self is to give it an identity that is outside our subjective experience of it. It does not have one. As magenta, the imaginary colors, and now this "Olo" that I desperately want to experience [8], colors are a multidimensional reperesention of a band of the electromagnetic spectrum, color is given a biological identity through our genes. In sense sense, the meaning of red can be understood as some proposition about the physical conditions of an evolutionary narrative that ends right now, describes the physical forces that caused the survival of our ancestors, and includes the the emergence of the channel rhodopsin gene that allows us to detect photons. So red doesn't possess a meaningful identity that is shared by the two of us, so I do not find "redness" to be a semantically useful distinction either.

What about the *roseness* of roses? It appears to be an external object, unlike red. It does not self-qualify like I do. But as we discuss roses, we find that we each have a canonical form of rose in our minds, and it is only by establishing the qualities of a rose and the relationships it shares to love, to gift-giving, and

to the absurd commitment to an upkeep-of-garden which give it the identity it has. It would seem that for me, *redness*, *roseness*, and *Dougness* all carry an intrinsic sense of identity that emerges from within. Best consider both to sit at the edge of meaning so as to reveal the shape of the semantic surface area occupied by the concept of *identity*.

We can begin to resolve this by asking a not-so-simple question: how do objects come to have identities at all, how do they come to have qualities, and how do they come to stand in relation to each other through the qualities they can, will, might, did, do, or do not share?

We pose an answer to that questions: the representation of the identity of objects and their qualities and the framework for representing their relationships, taken together, form a *Perspective*:

[Semantic] Definition 2. *A Perspective is a representational framework for identifying identities, qualifying qualities and referring to relationships [1].*

A formal definition of a Perspetive and a formal definition for constructing one, taken together, express the meaning of building a *coherent perspective* as a computational process.

Within the context of constructing a coherent perspective:

[Semantic] Assertion 3. *A Conscious Experience represents a change in Perspective. [1].*

[Colloquial] Example 1. *I catch you admiring a rose in my garden and I absentmindedly ask "what color is it?" You reply "Come over here and see for your*self*!"*

4.1 The Shape of a Semantic Surface

Hopefully the image of *semantic shape* is starting to form. Identity, identify, identification, quality, qualify, qualification, equality, reality, relation, related, relationship, representation, reference, reference frame, representational framework, perspective, perceptive, percieve, perception, expression, expressive, express, etc... These are not abstract notions in isolation. They are expressions of structure, rooted in a shared sense of semantics. A physical perspective isn't just a way of seeing. It is a referential framework that justifies *why* the things we see appear as they do, and *how* we recognize them as having meaning.

All throughout this discussion, a structure has been emerging: one in which identity arises through the qualification of qualities, and qualities inherit their meaning through a referential lineage. We've already used words like "referential framework," "intrinsic quality," and "self-evident identity." These are not just poetic phrases — they point toward a deeper semantic structure that is waiting to be formalized.

So let us take that step. If we want to express this semantic structures with integrity, so that we might articulate how identity, quality and relationship interact with referents precisely, staying grounded using concepts like the symbol-literal

dichotomy to keep the notion of identity semantically precise, we must define a language that is capable of expressing the definition of the concept of reference itself.

We have invented such a tool, and we call it *Semantic Reference Theory.*

5 Semantic Reference Theory

Self-reference is at the heart of a semantic comprehension of consciousness. Surely, the fact that we must refer to our *"self"* through our *"I"* gives a clue about how reference works in a semantic sense. A Gödel sentence did not come to articulate its own unprovability by containing itself. It's self-reference is expressed through talking about a sentence with a certain Gödel number which just so happens to be its own. We contend that this is how reference works semantically. There must be a symbol to serve as a locus for identity, to which qualities are applied before we can conclude that the objects they represent have certain properties.

We have assembled a collection of semantic assertions that express the nature of a *conscious experience*, with the goal of expressing it in a formalism in which it can be confirmed or falsified. We have developed a formalism to do exactly that. It's called Semantic Reference Theory. While it is a work in progress, it always will be by design. It is a formalism in which semantic forms about a certain topic are reflected upon in the context of related topics in order to cosntruct a formalism form them.

From one perspective, Semantic Reference Theory (SRT) is a tool to build Semantic Reference Theory. From another perspective, it can be expressed as an extension of first-order logic that provides structure on how it can be extended.

There is also a linguistic element to SRT. When one defines the semantics of a topic, we believe that you should choose a language most befitting of your topic. Of course, SRT must have a base form. It has a typographical lingo that gives it a unique flavor as a formalism. Representing reference symbolically is not unlike representing pointers in programming language. We chose a symbology with an aesthetic identity that uses programming as its inspiration.

It can be expressed as a first-order theory, and we ues thse linguistic constructs to replace the normal ones in formal logic:

!X means $\neg X$
X & Y means $X \wedge Y$
X | Y means $X \vee Y$
X =|> Y means $X \Rightarrow Y$
X <|=|> Y means $X \Leftrightarrow Y$
x == y means $x = y$
x != y means $x \neq y$
Ex: x == y means $\exists x : x = y$
Ax: x == y. means $\forall x : x = y$
AxEy: x == y means $\forall x \exists y : x = y$

The formalism of Semantic Reference Theory is founded on the essential dichotomy of representation: referents and their references. SRT is a first-order logic in the domain of referents. It has no constants, no functions, and only a single relation:

R(x,y,t)

This relation expresses that x relates to y through. We express this using our typographical domain specific language which is not itself a part of the formalism:

x =t> y

This syntax makes it clear that x is the subject of a relationship, y is the object of a relationship and t is a type of relationship. In this way, R(x,y,t) expresses the subject-object relationship as the fundamental semantic form.

The arrow syntax has expressive power. For instance, these are all valid expressions:

x =t> y means $R(x, y, t)$ is true
x !=t> y means $R(x, y, t)$ is not true
x <t= y means $R(y, x, t)$ is true
x <t=! y means $R(y, x, t)$ is not true
x <t=t> y means $R(x, y, t)$ and $R(y, x, t)$ are both true
x <t!=!t> y means $R(x, y, t)$ and $R(y, x, t)$ are both false

We will draw on definitions, axioms and theorems from various topics. For instance, the Semantics of Symbols expresses a symnbol in this way:

If the subject of the realtionship is the relationship itself, then x is a *symbol* for y and y is the *literal* of x:

x =x> y

We also make use of the representation function, which is a notational convenience. (y) deotes the canonical symbol for y. It satisfies the notion of a symbol entirely in terms of y:

(y) =(y)> y

The representation function is a notational convenience. One must be sure that y has a canonical system to make use of it.

A full treatment of Semantic Referenece Theory would be the topic of another paper, but we will provide links to the relavant topic as they are introduced into our reflections

6 The Semantics of Perspective

Insofar as Semantic Reference Theory exists to be a formal theory to capture the relationship between *language* and *meaning*, Perspective is one of its fundamental topics. It also happens to be essential for a theory of consciousness. The First-Person Perspective can't be formally defined without also formalizing the being that possesses one. Let's being our reflection here. Recall our definition of Perspective:

[Semantic] Definition 2. *A Perspective is a referential framework for qualifying identities, identifying qualities and expressing their relationships [1].*

Let's take this apart. First, we know that the framework is used for qualifying identities and identifying qualities. We also know that a Perspective belongs to an indentity. Let's denote that identity i, and call it the *qualifier* of the perspective.

Let us also imagine that the qualifier is one of a collection of object literals, none of which can reference anything. All reference must flow from a symbol that it literally represents. Therefore i istelf must have an identity. Let's call it (i).

For i to quality red, the following relation must hold:

(i) =red> (rose) <|=|> (rose) =red> rose

The first half of that sentence can be read as (i) identifying red as a quality of the rose, and the second half can be read as the expressing that the rose is red. Both qualities must be qualified by the qualifier.

How about identity? Perhaps we can consider identity a quality too. This statement would have to have meaning:

(i) =(rose)> (rose) <|=|> (rose) =(rose)> rose

This is a fascinating statement in Semantic Reference Theory. From The Semantics of Reference, we learn tha definition of a canonical. If the object of a relationship is the realtionship itself, then that object can bet taken to be the canonical value of a subject:

x =y> y

Form the expression above, you can see that for i to qualify the rose is to both give it an identity in the form of a symbol, and to take that identity as its canonical value. But this leads to an interesting definition:

[Semantic] Definition 2. *To* Identify *is to qualify an identity.*

Now, who qualifies the identity of i? As the one and only qualifier of the *Persepctive*, it must qualitfy itself:

(i) =(i)> (i) <|=|> (i) =(i)> i

What a curious phrase. On the Rightarrow, we see an expression that expresses the idea that the *self* belongs to the identity. On the left, we see a confluence of subject, object and relationship type.

From The Semantics of Equality we find that the notion of equality inherited from being a first-order theory is the only relationship type that can be used for direct self-reference:

x =eq> x

This means that for all x, R(x,x,eq) are the only true statements. By having the equality relationship be itself a refernt, this is a true proposition:

eq =eq> eq

From The Semantics of Relationship we find that relationship types have a notion of inheritance. Let -t> be the extension of =t> to ancentral relationships, defined as:

x -t> y <|=|> x =t> y | Ez: x -t> z & z -t> y

There exists a referent r such that:

x =t> y =|> t -r> r

This relationship repersents relationship itself. Since eq is the only relationship that admites x =x> x, then we can conclude:

(i) -r> eq

This reveals an important part of identity that we have overlooked. An identity seems to have a sense of itself. While I appear to be the one that qualifies the identity of both the rose and the color red, these entities seems to stand apart from me with a sense of themselves.

There is also a notion of a reference frame and its corresponding https://github.com/DNA-Platform/inexplicable-phenomena/formalism/semantic-reference-theory/reference.pdfframework. A framework is a series of relationship types that descend from a common on. If we reflect on this for a moment, and allow ourselves to imagine (i) as a referential framework for identity, we might be compelled to update our definition of qualification:

(i) =(red)> (red) <|=|> (red) =(red)> red & (red) -(i)> (i)

This establishes the idea that the red inherits its definiton from (i) as a relationship type in the framework that defines the perspective of (i). And inheriting from (i) as a relationship type requires:

(red) =(red)> (red) (apple) =(apple)> (apple)

Each one is a tiny equality operator, capable of asserting its own identity. Each thing you see from your perspective can be the subject and object of thought, and stands in relation to itself through you. And (i) is now a framework. Everything you see is defined in relaton to your identity.

There is an important distinction between the quality of red and the quality as intelligence. The quality of red is *self-evident*. The quality of intelligence is not. So let us make the distinction:

We qualify (red) as an identity through this expression:

(red) -(i)> (i)

The relationship ancestry can be direct or indirect. To identify red as an intrinsic quality is to say:

(red) =(i)> (i)

There is a way for a quality to be even more direct in the framework of (i). The intrinsic quality of red is expressed through the way (red) relates to (i). Well then the intrinsic quality of i is analogously expressed in the way (i) related to i:

(i) =(i)> (i)

That phrase is part of the broader expression of a qualifier that qualifies its own identity. To express the idea that the qualifier can identify the intrinsic quality of red is to say:

(i) =(red)> (red) <|=|> (red) =(red)> red & (red) =(i)> (i)

The full expression for the qualifier qualifying its own identity is:

(i) =(i)> (i) <|=|> (i) =(i)> i & (i) =(i)> (i)

You can see the redundancy on both sides of the expression. Self-identificaton and self-qualification was already an intrinsic quality of the qualifier. But to understand that possessing the quality of i is self-evident to the qualifier, and

that such a definition emergences naturally from the very notion of there being intrinsic qualities. It is plain to see that possessing a sense of self:

(i) =(i)> i

Is what it means for self-identity to to be an intrinsic quality. And intrinsic qualities are self-evident.

Example 1. You suddenly declare "I feel like a whole new man!" I respond in a concerned tone, "If you are not feeling like yourself today, you should consider seeing a doctor."

Let us go one step further and note that qualification takes thought. Intrinsic qualities are self-evident, but qualification itself is not.

Is it valid to qualify a rose as both rose as both red and blue?

(i) =red> (rose) & (i) =blue> (rose)

How about qualifying that you have two different ages?

(i) =[20-years-old]> (i) & (i) =[40-years-old]> (i)

Practiced qualifiers must apply consistent logic to have a coherent perspective, and that takes thought! So while thinking really hard about what it means to identify intrinsic qualities, and to qualify one's own identity, might I ask you to consider this expression:

((i) =(rose)> (rose) <|=|> (rose) =(rose)> rose & (rose) -(i)> (i)) &
((i) =red> (rose) <|=|> (rose) =red> rose & red =(i)> (i)) &
((i) =(i)> (i) <|=|> (i) =(i)> i & (i) =(i)> (i))

And ask yourself whether the last line qualifies for the interpretation:

"I think, therefore I am"?

It might help to ponder what it means to qualify someone as as a qualifier?

(i) =qualifier> (o) =|> (o) =(qualifier)> o

In the context of this reflection, being a qualifier means to possess a valid perspective. And not just any perspective! It's own perspective. Let's call that P(i). You may interpret that a P + (i) or as a function P with i as its argument. We can express that qualifying an object as a qualifier is to declare that they have their own perspective:

(i) =qualifier> (o) <|=|> (o) =(P(o))> o

To possess your own perspective is a wonderful way to express the semantics of being conscious:

(o) =(P(o))> o <|=|> (o) =(conscious)> o

So pleae consider the interpretation of this:

(qualifier) == (P(o)) == (conscious) == consciousness

6.1 An Unequivocally Inexplicable Transcendental Transformation

Transduction is an exotic concept with exotic usage. It has its etymological roots in the Latin nominative *trānsductiō*, which denotes the act or process of being *lead or carried across*. "ductiō" is an etymological ancestor of "educate." In this

sense, *transduction* seems to wish to mean a *transcendental transformation* of some sort.

Yet its meaning came to English in the form of the horizontal transfer of genetic material between bacteria, which is a sort of meta-genetic phenomenon in which a bacteriophage infects a bacterium, and with different degrees of symbiosis, captures fragments of its genome, after which they get *carried* in a capsid *across* the synaptic cleft of geneological ancestry, resulting in a novel genetic change and a novel form of genetic inheritance simultaneously. It strikes me as a form on genetic inheritance that is more like the genetic education you receive from your friends at school when you catch a cold after playing with them at recess.

I refer to it a *meta-genetic* because, while it may arise from through the process of cellular replication, it is not entailed by the replication process that is encoded in the genome of the cell. Were one to only believe in genetic inheritance through reproduction as expressed in the genome itself, the appearance of these new genes would be inexplicable. I offer up "Metagenetic Transduction" as a new term for horizontal gene transfer that more accurately captures the essence of its transcendantal nature.

In neuroscience, phototransduction refers to the process by which photons of light are *carried across* the rods and cones of our retina through absorbtion and converted into the electrochemical signals which to the action potential of the central nervous system. Similarly, mechanotransduction is when physical forces act on ion channels in our skin and other sensory organs to do basically the same thing. In both cases, physical stimuli elicit a response that has causal force within our nervous systems. These physical phenomena include photons of light, pressure waves of sound, pin pricks of pressurised pain, along with various other physical phenomena that correspond tactile and chemical transductive processes that kick-off the the sensation of touch, taste and smell. We happily shall, with gain of generality, refer to all of these phenomena as forms of *physical transduction*.

If a photon is the subject of a transduction, what is the object? What has that photon become? It would seem to have become a part of the fabric of a representational medium that reflects the properties of a photon, along with the properties of many other photons, along with the properties of many other physical phenomena all occuring at approximately the *same time*, based on a timescale defined by the pocketwatch of some being that experiences a tree to be more like a rock, and less like a bubble of boiling water, though it admits the properties of both at some timescale.

The corpus of physical stimuli, now having been *carried across* the synaptic cleft of actual synapses and into the land of physical phenomena that represent others, are carries along the river of representation until they reach its source, wherecy they are then absorbed into the event horizon of the epistemological black whole that we refer to as *the Hard Problem*. What comes out on the other end is the stuff that clocks, thoughts, decisions and dreams are made of.

We happen to know from the neuroscience literature that under the influence of powerful general anesthetics like propofol or isoflurane (cite), sensory input can still activate orientation pinwheels in the visual cortex, even in the complete absence of *perception*. All forms of conscious experience disappear completely as well, by virtue of semantics:

[Semantic] Assertion 1. *To perceive is to have conscious experiences. To have one is an act of perception. The act of perception is the definition of* having *a conscious experience [1].*

On the relationship between Conscious Experience and Perception, Chalmers notes: "Other experiences go along with perception in different modalities." So the object that he identifies as a conscious experience appears to be quantized according to the sense that represents it. He also points out that "the color and shape of a perceived object is integrated from separate visual pathways." There is tension here in the tense of the verb. Within a single sensory modality, does perception of the object happen first, at which point the qualities that correspond to its visual features are integrated? Or is *integration* fundamental to the act of perception?

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Acknowledgement It goes without saying: that *this* PDF, it's annotated *tex* file, the *html* version that it compiled to by way of markdown, and the evolutionary self-publishing Github repository that they all collectively self-reference... could only possibly be dedicated to the *author* of "Gödel, Escher, Bach," et al. Thank you for a lifetime of thought, Professor Hofstadter <3

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